

8. What happens when the population size changes? **Groups on the right-hand side of our classroom answer this one.**

We considered simple random samples drawn from a population of size 100,000. One might wonder: if the population is bigger, do we need to use larger samples to learn about it?

a. Make a graph of sampling distribution of sample proportion with **new population size 1,000,000** and sample size 50. Paste the graph here.

Fill in the blanks:

Simple random samples of size 50 from a population of size 1 million tend have 'Benefits' proportion around (fill in here).

Most samples have 'Benefits' proportion between (fill in) and (fill in).

Almost all samples have 'Benefits' proportion between (fill in) and (fill in).

The shape of the distribution of sample proportions is (fill in).

In the same way, one might wonder: if the population is smaller, do we learn much more about it with samples of size '50'?

b. Make a graph of the sampling distribution of sample proportions with **new population size 1000**, and sample size 50. Paste your graph here.

Fill in the blanks:

Simple random samples of size 50 from a population of size 1000 tend have 'Benefits' proportion around (fill in here).

Most samples have 'Benefits' proportion between (fill in) and (fill in).

Almost all samples have 'Benefits' proportion between (fill in) and (fill in).

The shape of the distribution of sample proportions is (fill in).

Intro to Sampling Distributions for Proportions

MATH 213

Group 1

Here are the first two questions from the R document.

1. Imagine that we choose a sample of people from our population and ask people in the sample whether they think that the work scientists do benefits people like them. What population parameter could we use the results of our poll to estimate? What statistic would we calculate to make this estimate?

Parameter: **the percentage of the population that think scientists do benefit them**

Statistic: **the percentage of the sample that think scientists do benefit them**

2. If we want to get an unbiased estimate of the population parameter, how should we gather our sample? (For the moment, think about the simplest strategy that we have discussed for creating a representative sample.)

Simple random sampling

3. Use the following template to describe the class stacked dot plot.

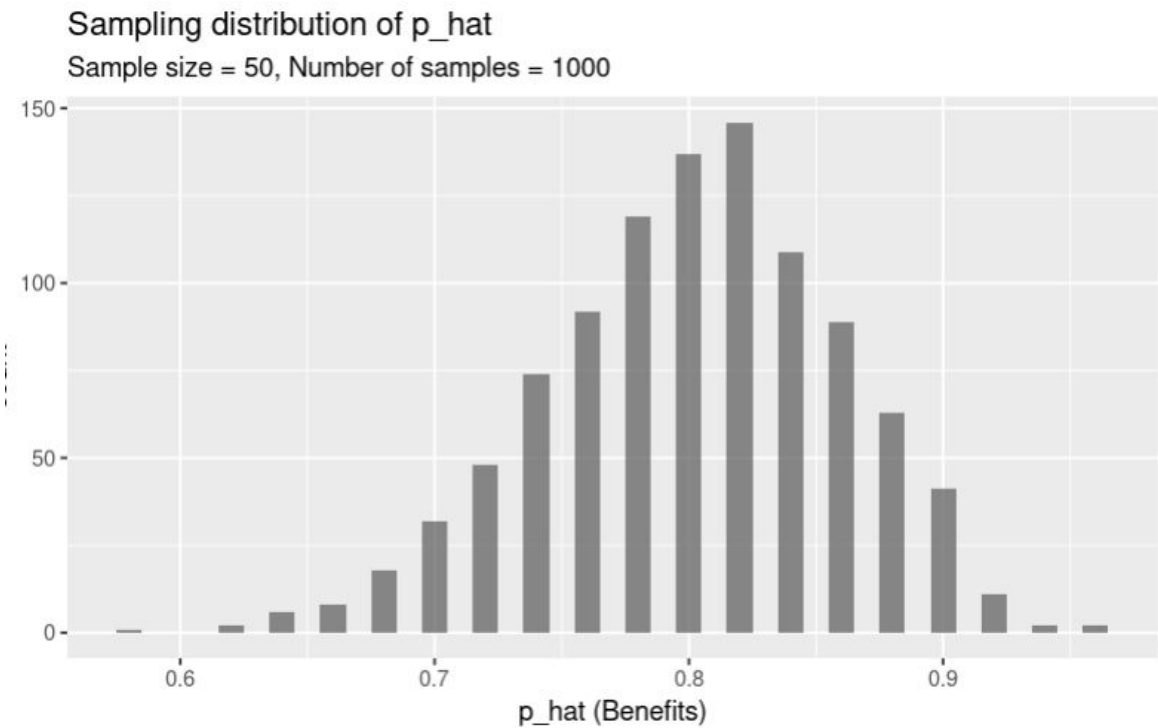
Simple random samples of size 50 tend have `Benefits` proportion around 82

Most samples have `Benefits` proportion between 75 and 85.

Almost all samples have `Benefits` proportion between 70 and 90.

The shape of the distribution of sample proportions is slightly. Skewed left

4. Put your histogram of the sampling distribution here.



5. Record the mean, standard error, Q1, median, and Q3 for your histogram here.

Mean: 0.8013

SE: .58

Q1: 0.76

Median: .8

6. Use the following template to describe the sampling distribution. Use your histogram and the summary statistics from the previous slide.

Simple random samples of size 50 tend have `Benefits` proportion around **80**

Most samples have `Benefits` proportion between **.76** and **.84**

Almost all samples have `Benefits` proportion between and **.58** and **.96**

The shape of the distribution of sample proportions is centered at **.8 with a slight right skew**

7. What happens when the sample size changes? **Groups on the left-hand side of our classroom answer this one. (If you are on the right-hand side, see the next slide.)**

a. Create the sampling distribution with samples of size `10` instead of size `50`. Paste your graph below.

Fill in the blanks:

Simple random samples of size 10 tend have `Benefits` proportion around.

Most samples have `Benefits` proportion between and.

Almost all samples have `Benefits` proportion between and.

The shape of the distribution of sample proportions is

b. Create the sampling distribution with samples of size `500` instead of size `50`. Paste your graph below.

Fill in the blanks:

Simple random samples of size 500 tend have `Benefits` proportion around.

Most samples have `Benefits` proportion between and

Almost all samples have `Benefits` proportion between and.

The shape of the distribution of sample proportions is

8. What happens when the population size changes? **Groups on the right-hand side of our classroom answer this one.**

We considered simple random samples drawn from a population of size 100,000. One might wonder: if the population is bigger, do we need to use larger samples to learn about it?

a. Make a graph of sampling distribution of sample proportion with **new population size 1,000,000** and sample size 50. Paste the graph here.

Fill in the blanks:

Simple random samples of size 50 from a population of size 1 million tend have `Benefits` proportion around (fill in here).

Most samples have `Benefits` proportion between (fill in) and (fill in).

Almost all samples have `Benefits` proportion between (fill in) and (fill in).

The shape of the distribution of sample proportions is (fill in).

In the same way, one might wonder: if the population is smaller, do we learn much more about it with samples of size `50`?

b. Make a graph of the sampling distribution of sample proportions with **new population size 1000**, and sample size 50. Paste your graph here.

Fill in the blanks:

Simple random samples of size 50 from a population of size 1000 tend have `Benefits` proportion around (fill in here).

Most samples have `Benefits` proportion between (fill in) and (fill in).

Almost all samples have `Benefits` proportion between (fill in) and (fill in).

The shape of the distribution of sample proportions is (fill in).

Group 2

Here are the first two questions from the R document.

1. Imagine that we choose a sample of people from our population and ask people in the sample whether they think that the work scientists do benefits people like them. What population parameter could we use the results of our poll to estimate? What statistic would we calculate to make this estimate?

Parameter: Percentage of the population that thinks the work scientists do benefits people like them

Statistic: Percentage of the sample that thinks the work scientists do benefits people like them

2. If we want to get an unbiased estimate of the population parameter, how should we gather our sample? (For the moment, think about the simplest strategy that we have discussed for creating a representative sample.)

Simple random sampling

3. Use the following template to describe the class stacked dot plot.

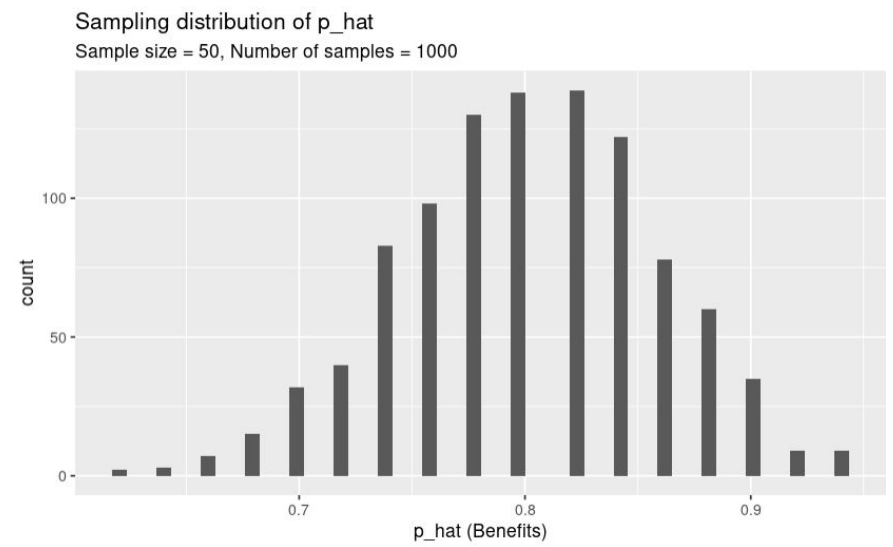
Simple random samples of size 50 tend have `Benefits` proportion around 82

Most samples have `Benefits` proportion between 75 and 87.

Almost all samples have `Benefits` proportion between 70 and 90.

The shape of the distribution of sample proportions is slightly left skewed.

4. Put your histogram of the sampling distribution here.



5. Record the mean, standard error, Q1, median, and Q3 for your histogram here.

scientist_work<chr>	sample_mean<dbl>	sample_sd<dbl>	sample_q1<dbl>	sample_median<dbl>	sample_q3<dbl>
Benefits	0.80146	0.05613394	0.76	0.8	0.84

6. Use the following template to describe the sampling distribution. Use your histogram and the summary statistics from the previous slide.

Simple random samples of size 50 tend have `Benefits` proportion around 80.

Most samples have `Benefits` proportion between 76 and 84.

Almost all samples have `Benefits` proportion between 68 and 90.

The shape of the distribution of sample proportions is centered around 80/0.8 with a slight left skew.

A tibble: 1 x 8

scientist_work <chr>	sample_mean <dbl>	sample_sd <dbl>	sample_q025 <dbl>	sample_q25 <dbl>	sample_median <dbl>	sample_q75 <dbl>	sample_q975 <dbl>
Benefits	0.80146	0.05613394	0.68	0.76	0.8	0.84	0.9

1 row

7. What happens when the sample size changes? **Groups on the left-hand side of our classroom answer this one. (If you are on the right-hand side, see the next slide.)**

a. Create the sampling distribution with samples of size `10` instead of size `50`. Paste your graph below.

Fill in the blanks:

Simple random samples of size 10 tend have `Benefits` proportion around.

Most samples have `Benefits` proportion between and.

Almost all samples have `Benefits` proportion between and.

The shape of the distribution of sample proportions is

b. Create the sampling distribution with samples of size `500` instead of size `50`. Paste your graph below.

Fill in the blanks:

Simple random samples of size 500 tend have `Benefits` proportion around.

Most samples have `Benefits` proportion between and

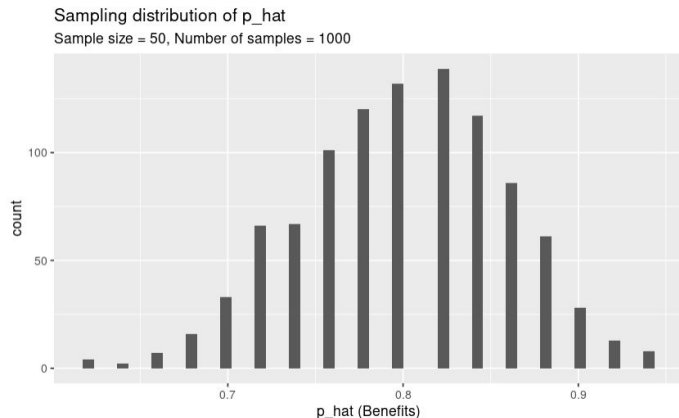
Almost all samples have `Benefits` proportion between and.

The shape of the distribution of sample proportions is

8. What happens when the population size changes? **Groups on the right-hand side of our classroom answer this one.**

We considered simple random samples drawn from a population of size 100,000. One might wonder: if the population is bigger, do we need to use larger samples to learn about it?

a. Make a graph of sampling distribution of sample proportion with **new population size 1,000,000** and sample size 50. Paste the graph here.



Fill in the blanks:

Simple random samples of size 50 from a population of size 1 million tend have `Benefits` proportion around (80).

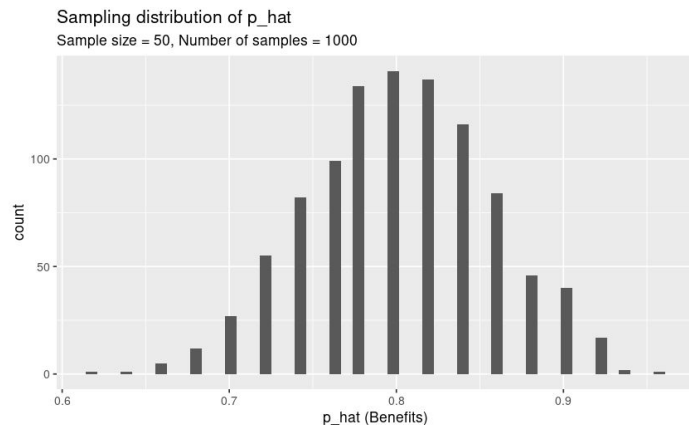
Most samples have `Benefits` proportion between (76) and (84).

Almost all samples have `Benefits` proportion between (68) and (90).

The shape of the distribution of sample proportions is (slightly left skewed).

In the same way, one might wonder: if the population is smaller, do we learn much more about it with samples of size `50`?

b. Make a graph of the sampling distribution of sample proportions with **new population size 1000**, and sample size 50. Paste your graph here.



Fill in the blanks:

Simple random samples of size 50 from a population of size 1000 tend have `Benefits` proportion around (80).

Most samples have `Benefits` proportion between (76) and (84).

Almost all samples have `Benefits` proportion between (70) and (90).

The shape of the distribution of sample proportions is (slightly left skewed).

Group 3

Here are the first two questions from the R document.

1. Imagine that we choose a sample of people from our population and ask people in the sample whether they think that the work scientists do benefits people like them. What population parameter could we use the results of our poll to estimate? What statistic would we calculate to make this estimate?

Parameter: Percentage of the population that think scientist work do benefit them.

Statistic: Percentage of the sample that thinks scientists work do benefit them.

2. If we want to get an unbiased estimate of the population parameter, how should we gather our sample? (For the moment, think about the simplest strategy that we have discussed for creating a representative sample.)

Assign everyone a number and use a random number gen to get the participants.

3. Use the following template to describe the class stacked dot plot.

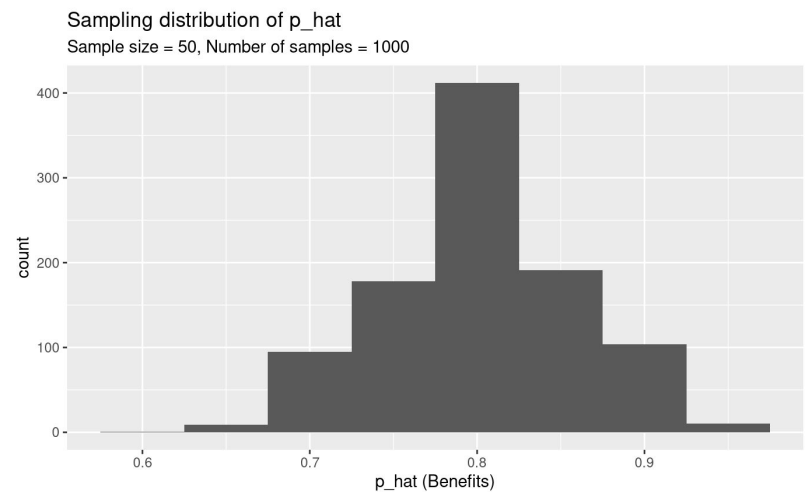
Simple random samples of size 50 tend have `Benefits` proportion around 80 %

Most samples have `Benefits` proportion between and .
76 to 86%

Almost all samples have `Benefits` proportion between and .
60% and 90%

The shape of the distribution of sample proportions is...
The proportion of the graph is slightly left skewed.

4. Put your histogram of the sampling distribution here.



5. Record the mean, standard error, Q1, median, and Q3 for your histogram here.

Mean - .80

Standard error- 0.00176

Q1 - 0.76

Median - 0.8

Q3 - 0.84

6. Use the following template to describe the sampling distribution. Use your histogram and the summary statistics from the previous slide.

Simple random samples of size 50 tend have `Benefits` proportion around.
0.8

Most samples have `Benefits` proportion between and.
0.75 - 0.85

Almost all samples have `Benefits` proportion between and.
0.65 and 0.95

The shape of the distribution of sample proportions is .
Proportional

7. What happens when the sample size changes? **Groups on the left-hand side of our classroom answer this one. (If you are on the right-hand side, see the next slide.)**

a. Create the sampling distribution with samples of size `10` instead of size `50`. Paste your graph below.

Fill in the blanks:

Simple random samples of size 10 tend have `Benefits` proportion around.

Most samples have `Benefits` proportion between and.

Almost all samples have `Benefits` proportion between and.

The shape of the distribution of sample proportions is

b. Create the sampling distribution with samples of size `500` instead of size `50`. Paste your graph below.

Fill in the blanks:

Simple random samples of size 500 tend have `Benefits` proportion around.

Most samples have `Benefits` proportion between and

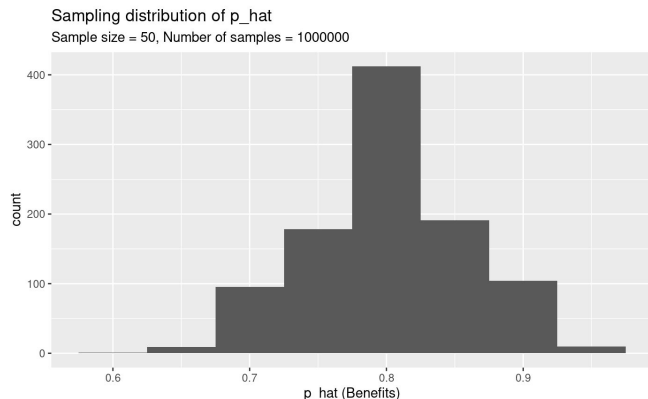
Almost all samples have `Benefits` proportion between and.

The shape of the distribution of sample proportions is

8. What happens when the population size changes? **Groups on the right-hand side of our classroom answer this one.**

We considered simple random samples drawn from a population of size 100,000. One might wonder: if the population is bigger, do we need to use larger samples to learn about it?

a. Make a graph of sampling distribution of sample proportion with **new population size 1,000,000** and sample size 50. Paste the graph here.



Fill in the blanks:

Simple random samples of size 50 from a population of size 1 million tend have `Benefits` proportion around 0.8, or 80%..

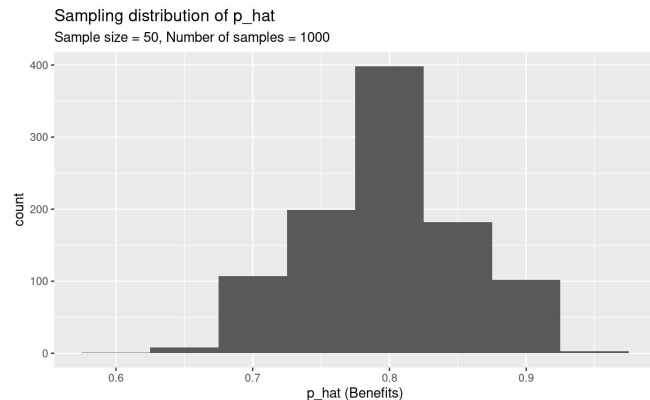
Most samples have `Benefits` proportion between 0.75 and 0.85.

Almost all samples have `Benefits` proportion between 0.65 and 0.95.

The shape of the distribution of sample proportions is bell shaped.

In the same way, one might wonder: if the population is smaller, do we learn much more about it with samples of size `50`?

b. Make a graph of the sampling distribution of sample proportions with **new population size 1000**, and sample size 50. Paste your graph here.



Fill in the blanks:

Simple random samples of size 50 from a population of size 1000 tend have `Benefits` proportion around 0.8.

Most samples have `Benefits` proportion between 0.75 and 0.85.

Almost all samples have `Benefits` proportion between 0.65 and 0.95.

The shape of the distribution of sample proportions is bell shaped..

Group 4

Here are the first two questions from the R document.

1. Imagine that we choose a sample of people from our population and ask people in the sample whether they think that the work scientists do benefits people like them. What population parameter could we use the results of our poll to estimate? What statistic would we calculate to make this estimate?

Parameter: The proportion of the 100,000 adults who think that the work of scientists benefit people like them

Statistic: The proportion of the adults polled who think that the work of scientists benefit people like them

2. If we want to get an unbiased estimate of the population parameter, how should we gather our sample? (For the moment, think about the simplest strategy that we have discussed for creating a representative sample.)

Simple random sampling

3. Use the following template to describe the class stacked dot plot.

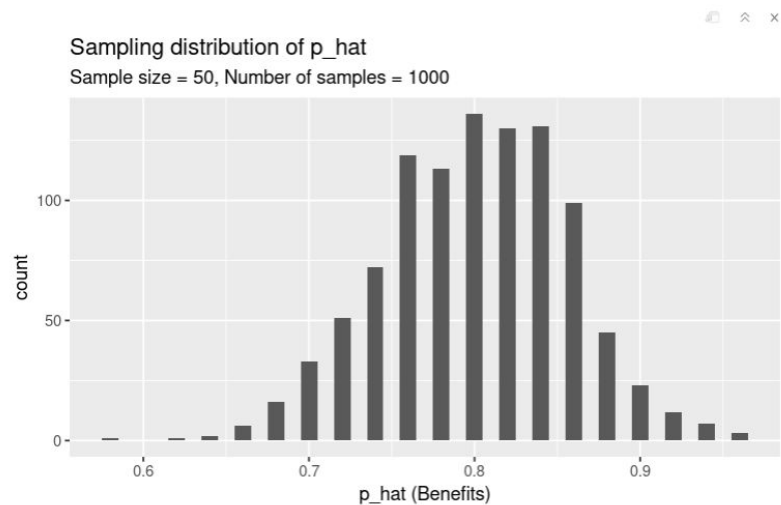
Simple random samples of size 50 tend have `Benefits` proportion around 0.8

Most samples have `Benefits` proportion between 0.78 and 0.86.

Almost all samples have `Benefits` proportion between .74 and .90.

The shape of the distribution of sample proportions is slightly left skewed.

4. Put your histogram of the sampling distribution here.



5. Record the mean, standard error, Q1, median, and Q3 for your histogram here.

A tibble: 1 × 8

scientist_work<chr>	mean<dbl>	median<dbl>	sd<dbl>	q025<dbl>	q25<dbl>	q75<dbl>	q975<dbl>
Benefits	0.80038	0.8	0.0561873	0.68	0.76	0.84	0.9

1 row

6. Use the following template to describe the sampling distribution. Use your histogram and the summary statistics from the previous slide.

Simple random samples of size 50 tend have `Benefits` proportion around 0.8.

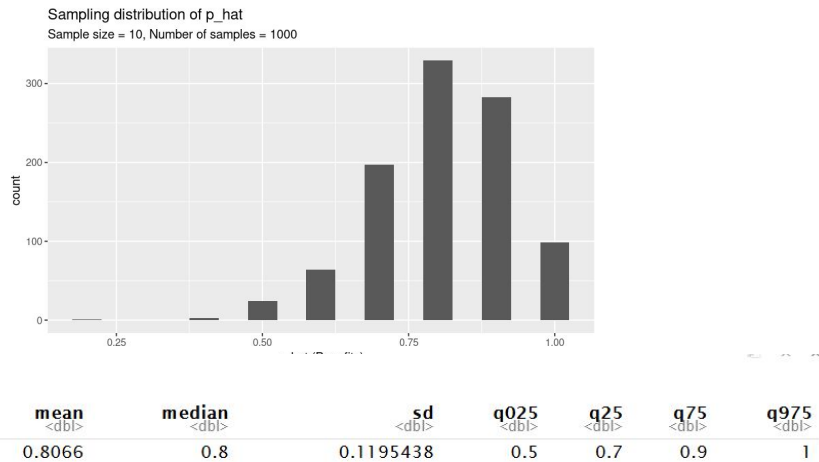
Most samples have `Benefits` proportion between 0.76 and 0.84.

Almost all samples have `Benefits` proportion between 0.68 and 0.90.

The shape of the distribution of sample proportions is normal.

7. What happens when the sample size changes? **Groups on the left-hand side of our classroom answer this one. (If you are on the right-hand side, see the next slide.)**

- a. Create the sampling distribution with samples of size `10` instead of size `50`. Paste your graph below.



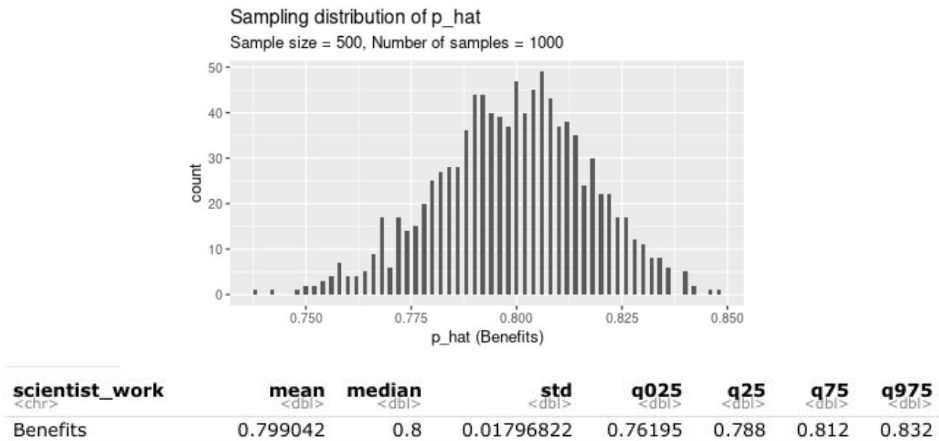
Simple random samples of size 10 tend have `Benefits` proportion around .8.

Most samples have `Benefits` proportion between .7 and .9.

Almost all samples have `Benefits` proportion between .5 and 1.

The shape of the distribution of sample proportions is left skewed

- b. Create the sampling distribution with samples of size `500` instead of size `50`. Paste your graph below.



Fill in the blanks:

Simple random samples of size 500 tend have `Benefits` proportion around 0.8.

Most samples have `Benefits` proportion between 0.79 and 0.81.

Almost all samples have `Benefits` proportion between 0.76 and 0.83.

The shape of the distribution of sample proportions is normal

8. What happens when the population size changes? **Groups on the right-hand side of our classroom answer this one.**

We considered simple random samples drawn from a population of size 100,000. One might wonder: if the population is bigger, do we need to use larger samples to learn about it?

a. Make a graph of sampling distribution of sample proportion with **new population size 1,000,000** and sample size 50. Paste the graph here.

Fill in the blanks:

Simple random samples of size 50 from a population of size 1 million tend have `Benefits` proportion around (fill in here).

Most samples have `Benefits` proportion between (fill in) and (fill in).

Almost all samples have `Benefits` proportion between (fill in) and (fill in).

The shape of the distribution of sample proportions is (fill in).

In the same way, one might wonder: if the population is smaller, do we learn much more about it with samples of size `50`?

b. Make a graph of the sampling distribution of sample proportions with **new population size 1000**, and sample size 50. Paste your graph here.

Fill in the blanks:

Simple random samples of size 50 from a population of size 1000 tend have `Benefits` proportion around (fill in here).

Most samples have `Benefits` proportion between (fill in) and (fill in).

Almost all samples have `Benefits` proportion between (fill in) and (fill in).

The shape of the distribution of sample proportions is (fill in).

Group 5

Here are the first two questions from the R document.

1. Imagine that we choose a sample of people from our population and ask people in the sample whether they think that the work scientists do benefits people like them. What population parameter could we use the results of our poll to estimate? What statistic would we calculate to make this estimate?

Parameter: **Proportion of people globally who believe the work scientists do benefits them**

Statistic: **Percentage of people in the sample who believe the work of scientists benefits them.**

2. If we want to get an unbiased estimate of the population parameter, how should we gather our sample? (For the moment, think about the simplest strategy that we have discussed for creating a representative sample.)

Randomly polling people in our chosen sample of the population

3. Use the following template to describe the class stacked dot plot.

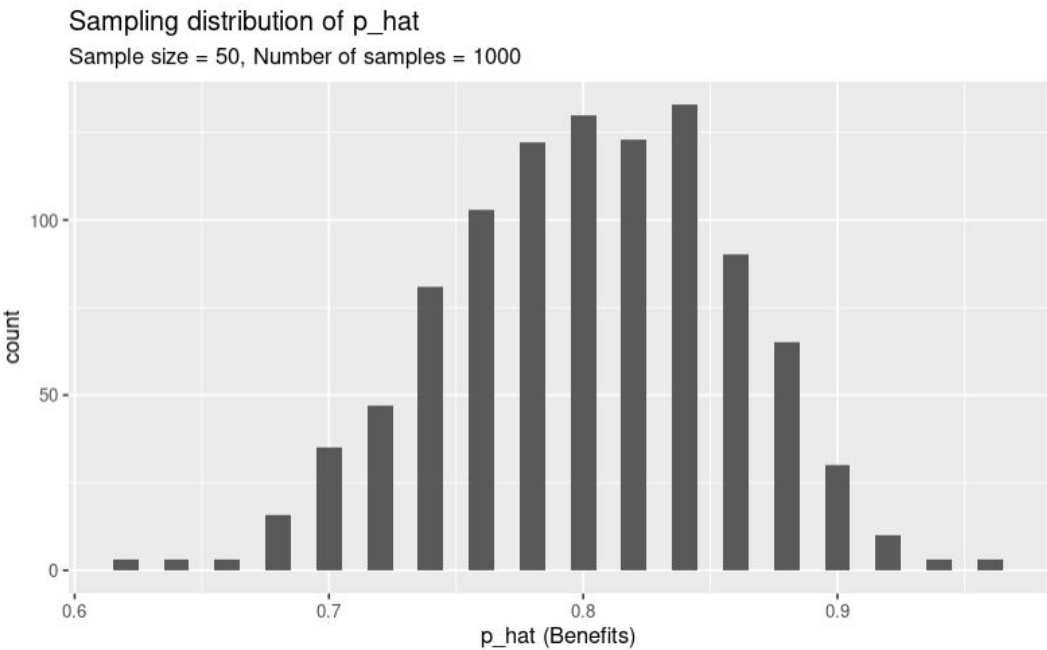
Simple random samples of size 50 tend to have `Benefits` proportion around 0.82

Most samples have `Benefits` proportion between 0.78 and 0.82 .

Almost all samples have `Benefits` proportion between 0.74 and 0.87 .

The shape of the distribution of sample proportions is slightly left skewed.

4. Put your histogram of the sampling distribution



5. Record the mean, standard error, Q1, median, and Q3 for your histogram here.

Mean: 0.80144
Standard error: 0.05658558
Q1: 0.76
Median: 0.8
Q3: 0.84

6. Use the following template to describe the sampling distribution. Use your histogram and the summary statistics from the previous slide.

Simple random samples of size 50 tend have `Benefits` proportion around 0.85.

Most samples have `Benefits` proportion between 0.72 and 0.90

Almost all samples have `Benefits` proportion between 0.68 and 0.88.

The shape of the distribution of sample proportions is mostly normal .

7. What happens when the sample size changes? **Groups on the left-hand side of our classroom answer this one. (If you are on the right-hand side, see the next slide.)**

a. Create the sampling distribution with samples of size `10` instead of size `50`. Paste your graph below.

Fill in the blanks:

Simple random samples of size 10 tend have `Benefits` proportion around.

Most samples have `Benefits` proportion between and.

Almost all samples have `Benefits` proportion between and.

The shape of the distribution of sample proportions is

b. Create the sampling distribution with samples of size `500` instead of size `50`. Paste your graph below.

Fill in the blanks:

Simple random samples of size 500 tend have `Benefits` proportion around.

Most samples have `Benefits` proportion between and

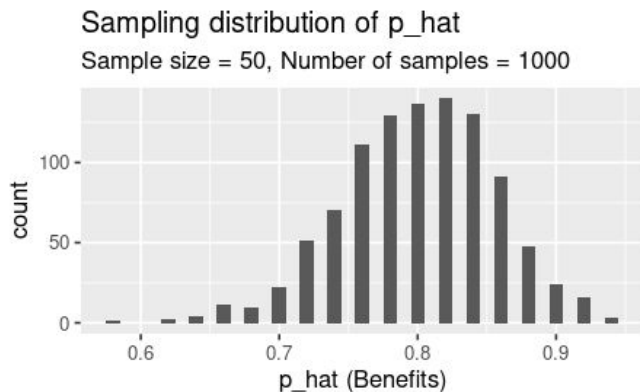
Almost all samples have `Benefits` proportion between and.

The shape of the distribution of sample proportions is

8. What happens when the population size changes? **Groups on the right-hand side of our classroom answer this one.**

We considered simple random samples drawn from a population of size 100,000. One might wonder: if the population is bigger, do we need to use larger samples to learn about it?

a. Make a graph of sampling distribution of sample proportion with **new population size 1,000,000** and sample size 50. Paste the graph here.



Fill in the blanks:

Simple random samples of size 50 from a population of size 1 million tend have `Benefits` proportion around 0.80.

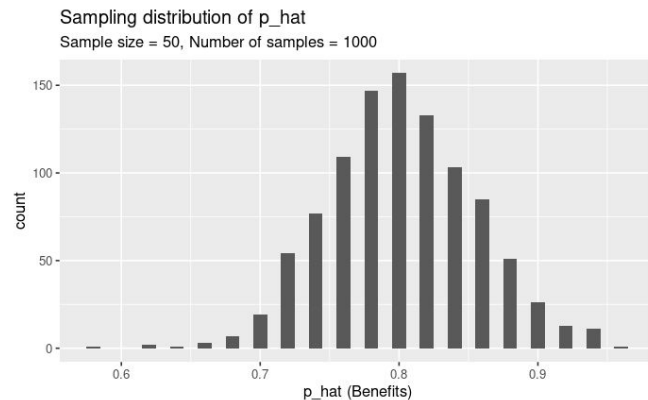
Most samples have `Benefits` proportion between 0.76 and 0.84

Almost all samples have `Benefits` proportion between 0.68 and 0.90.

The shape of the distribution of sample proportions is normal

In the same way, one might wonder: if the population is smaller, do we learn much more about it with samples of size `50`?

b. Make a graph of the sampling distribution of sample proportions with **new population size 1000**, and sample size 50. Paste your graph here.



Fill in the blanks:

Simple random samples of size 50 from a population of size 1000 tend have `Benefits` proportion around 0.8.

Most samples have `Benefits` proportion between 0.76 and 0.84.

Almost all samples have `Benefits` proportion between 0.7 and 0.9

The shape of the distribution of sample proportions is normal.

Group 6

Here are the first two questions from the R document.

1. Imagine that we choose a sample of people from our population and ask people in the sample whether they think that the work scientists do benefits people like them. What population parameter could we use the results of our poll to estimate? What statistic would we calculate to make this estimate?

Parameter: All adults in Frederick County, Maryland that believe scientists do work to benefit people like them.

Statistic: Percentage of those asked in the sample that believe scientists do work to benefit people like them.

2. If we want to get an unbiased estimate of the population parameter, how should we gather our sample? (For the moment, think about the simplest strategy that we have discussed for creating a representative sample.)

We should gather a large, randomized sample in order to get the most accurate estimate of the population.

3. Use the following template to describe the class stacked dot plot.

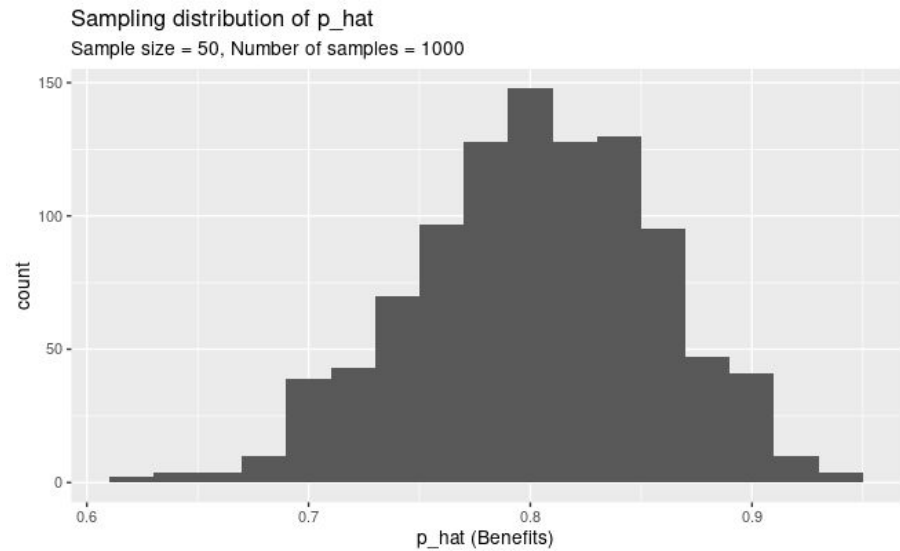
Simple random samples of size 50 tend have `Benefits` proportion around **80%**

Most samples have `Benefits` proportion between **76%** and **86%**.

Almost all samples have `Benefits` proportion between **71%** and **90%**.

The shape of the distribution of sample proportions is slightly **bell curved**.

4. Put your histogram of the sampling distribution here.



5. Record the mean, standard error, Q1, median, and Q3 for your histogram here.

scientist_work<chr>	mean_p_hat<dbl>	standard_error<dbl>	Q1<dbl>	median<dbl>	Q3<dbl>
Benefits	0.80234	0.05518306	0.76	0.8	0.84

6. Use the following template to describe the sampling distribution. Use your histogram and the summary statistics from the previous slide.

Simple random samples of size 50 tend have `Benefits` proportion around **0.802**.

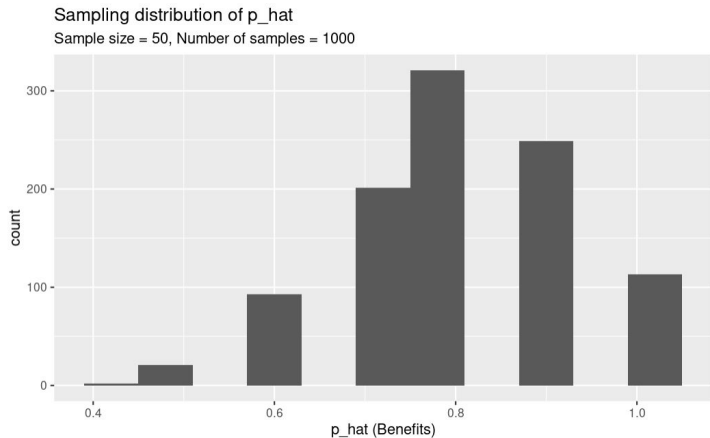
Most samples have `Benefits` proportion between **0.76** and **0.84**.

Almost all samples have `Benefits` proportion between **0.70** and **0.90**.

The shape of the distribution of sample proportions is **bell shaped**.

7. What happens when the sample size changes? **Groups on the left-hand side of our classroom answer this one. (If you are on the right-hand side, see the next slide.)**

- a. Create the sampling distribution with samples of size `10` instead of size `50`. Paste your graph below.



Fill in the blanks:

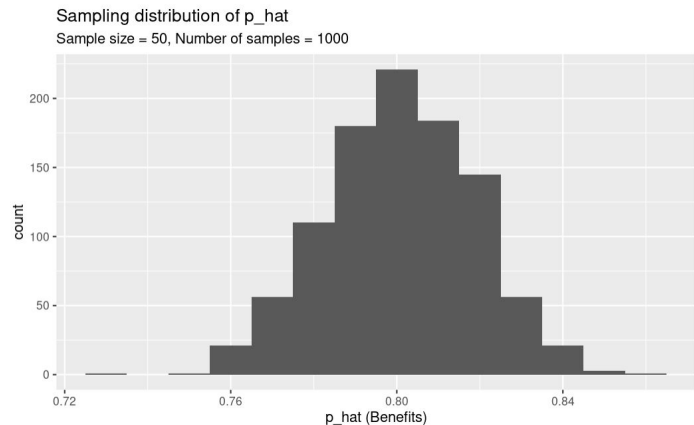
Simple random samples of size 10 tend have `Benefits` proportion around **0.8**.

Most samples have `Benefits` proportion between **0.7** and **0.9**.

Almost all samples have `Benefits` proportion between **0.6** and 1.

The shape of the distribution of sample proportions is **bell shaped**.

- b. Create the sampling distribution with samples of size `500` instead of size `50`. Paste your graph below.



Fill in the blanks:

Simple random samples of size 500 tend have `Benefits` proportion around **0.802**.

Most samples have `Benefits` proportion between **0.788** and **0.814**

Almost all samples have `Benefits` proportion between **0.766** and **0.834**.

The shape of the distribution of sample proportions is **bell shaped**.

8. What happens when the population size changes? **Groups on the right-hand side of our classroom answer this one.**

We considered simple random samples drawn from a population of size 100,000. One might wonder: if the population is bigger, do we need to use larger samples to learn about it?

a. Make a graph of sampling distribution of sample proportion with **new population size 1,000,000** and sample size 50. Paste the graph here.

Fill in the blanks:

Simple random samples of size 50 from a population of size 1 million tend have `Benefits` proportion around (fill in here).

Most samples have `Benefits` proportion between (fill in) and (fill in).

Almost all samples have `Benefits` proportion between (fill in) and (fill in).

The shape of the distribution of sample proportions is (fill in).

In the same way, one might wonder: if the population is smaller, do we learn much more about it with samples of size `50`?

b. Make a graph of the sampling distribution of sample proportions with **new population size 1000**, and sample size 50. Paste your graph here.

Fill in the blanks:

Simple random samples of size 50 from a population of size 1000 tend have `Benefits` proportion around (fill in here).

Most samples have `Benefits` proportion between (fill in) and (fill in).

Almost all samples have `Benefits` proportion between (fill in) and (fill in).

The shape of the distribution of sample proportions is (fill in).

Group 7

Here are the first two questions from the R document.

1. Imagine that we choose a sample of people from our population and ask people in the sample whether they think that the work scientists do benefits people like them. What population parameter could we use the results of our poll to estimate? What statistic would we calculate to make this estimate?

Parameter: P

Statistic: proportion of people who don't believe science affects them in the population

2. If we want to get an unbiased estimate of the population parameter, how should we gather our sample? (For the moment, think about the simplest strategy that we have discussed for creating a representative sample.)

Grab a list of all frederick county adults, randomly select 1000. Include a representative proportion of people from all demographics such as socioeconomic status, race, age, education...

3. Use the following template to describe the class stacked dot plot.

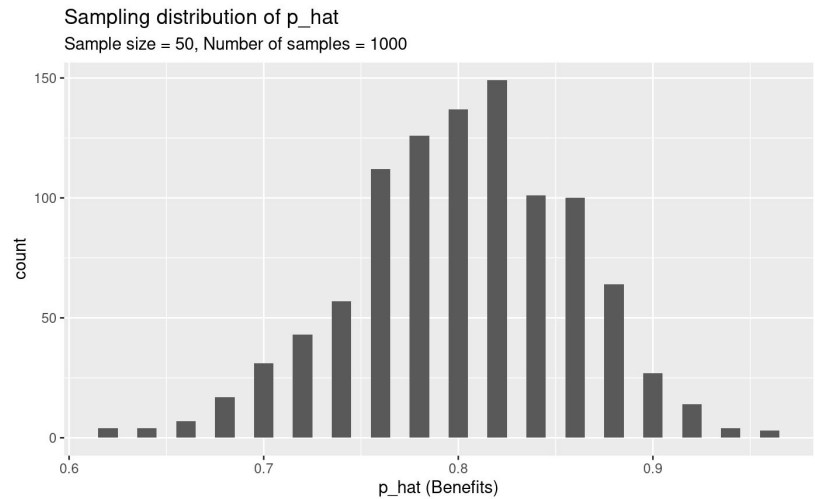
Simple random samples of size 50 tend have `Benefits` proportion around 0.86 - 0.9

Most samples have `Benefits` proportion between and 0.79 - 0.9.

Almost all samples have `Benefits` proportion between and .
0.7 - 0.92

The shape of the distribution of sample proportions is slightly.
Left skewed

4. Put your histogram of the sampling distribution here.



5. Record the mean, standard error, Q1, median, and Q3 for your histogram here.

Mean:

Standard error:

Q1:

Median:

Q3:

6. Use the following template to describe the sampling distribution. Use your histogram and the summary statistics from the previous slide.

Simple random samples of size 50 tend have `Benefits` proportion around.

Most samples have `Benefits` proportion between and.

Almost all samples have `Benefits` proportion between and.

The shape of the distribution of sample proportions is .

The sample distribution has a shape that is symmetrical and centered around 0.8 with a spread of around 0.6 to just over 0.9. Most samples have benefits around 0.7 to 0.9 and all have benefits from ().

7. What happens when the sample size changes? **Groups on the left-hand side of our classroom answer this one. (If you are on the right-hand side, see the next slide.)**

a. Create the sampling distribution with samples of size `10` instead of size `50`. Paste your graph below.

Fill in the blanks:

Simple random samples of size 10 tend have `Benefits` proportion around.

Most samples have `Benefits` proportion between and.

Almost all samples have `Benefits` proportion between and.

The shape of the distribution of sample proportions is

b. Create the sampling distribution with samples of size `500` instead of size `50`. Paste your graph below.

Fill in the blanks:

Simple random samples of size 500 tend have `Benefits` proportion around.

Most samples have `Benefits` proportion between and

Almost all samples have `Benefits` proportion between and.

The shape of the distribution of sample proportions is

8. What happens when the population size changes? **Groups on the right-hand side of our classroom answer this one.**

We considered simple random samples drawn from a population of size 100,000. One might wonder: if the population is bigger, do we need to use larger samples to learn about it?

a. Make a graph of sampling distribution of sample proportion with **new population size 1,000,000** and sample size 50. Paste the graph here.

Fill in the blanks:

Simple random samples of size 50 from a population of size 1 million tend have 'Benefits' proportion around (fill in here).

Most samples have 'Benefits' proportion between (fill in) and (fill in).

Almost all samples have 'Benefits' proportion between (fill in) and (fill in).

The shape of the distribution of sample proportions is (fill in).

In the same way, one might wonder: if the population is smaller, do we learn much more about it with samples of size '50'?

b. Make a graph of the sampling distribution of sample proportions with **new population size 1000**, and sample size 50. Paste your graph here.

Fill in the blanks:

Simple random samples of size 50 from a population of size 1000 tend have 'Benefits' proportion around (fill in here).

Most samples have 'Benefits' proportion between (fill in) and (fill in).

Almost all samples have 'Benefits' proportion between (fill in) and (fill in).

The shape of the distribution of sample proportions is (fill in).

Group 8

Here are the first two questions from the R document.

1. Imagine that we choose a sample of people from our population and ask people in the sample whether they think that the work scientists do benefits people like them. What population parameter could we use the results of our poll to estimate? What statistic would we calculate to make this estimate?

Parameter: **The proportion of people in the population who believe that the work scientists do benefits people like them.**

Statistic: **We can calculate this estimate with the 80000 people in the population who believe they benefit.**

2. If we want to get an unbiased estimate of the population parameter, how should we gather our sample? (For the moment, think about the simplest strategy that we have discussed for creating a representative sample.)

The most unbiased sampling method would be a blind random sampling of people

3. Use the following template to describe the class stacked dot plot.

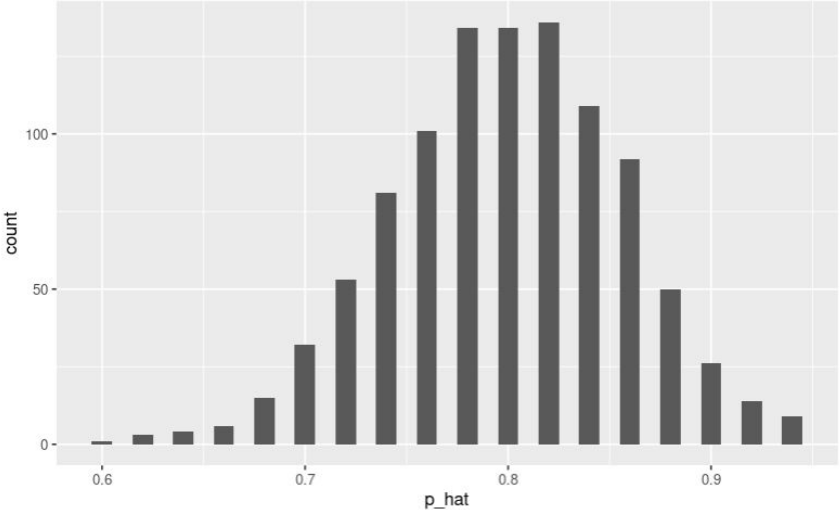
Simple random samples of size 50 tend have `Benefits` proportion around 0.80

Most samples have `Benefits` proportion between 75 and 90

Almost all samples have `Benefits` proportion between 70 and 95.

The shape of the distribution of sample proportions is slightly right skewed.

4. Put your histogram of the sampling distribution here.



5. Record the mean, standard error, Q1, median, and Q3 for your histogram here.

Mean:

Standard error:

Q1:

Median:

Q3:

6. Use the following template to describe the sampling distribution. Use your histogram and the summary statistics from the previous slide.

Simple random samples of size 50 tend have `Benefits` proportion around.

Most samples have `Benefits` proportion between and.

Almost all samples have `Benefits` proportion between and.

The shape of the distribution of sample proportions is .

7. What happens when the sample size changes? **Groups on the left-hand side of our classroom answer this one. (If you are on the right-hand side, see the next slide.)**

a. Create the sampling distribution with samples of size `10` instead of size `50`. Paste your graph below.

Fill in the blanks:

Simple random samples of size 10 tend have `Benefits` proportion around.

Most samples have `Benefits` proportion between and.

Almost all samples have `Benefits` proportion between and.

The shape of the distribution of sample proportions is

b. Create the sampling distribution with samples of size `500` instead of size `50`. Paste your graph below.

Fill in the blanks:

Simple random samples of size 500 tend have `Benefits` proportion around.

Most samples have `Benefits` proportion between and

Almost all samples have `Benefits` proportion between and.

The shape of the distribution of sample proportions is

8. What happens when the population size changes? **Groups on the right-hand side of our classroom answer this one.**

We considered simple random samples drawn from a population of size 100,000. One might wonder: if the population is bigger, do we need to use larger samples to learn about it?

a. Make a graph of sampling distribution of sample proportion with **new population size 1,000,000** and sample size 50. Paste the graph here.

Fill in the blanks:

Simple random samples of size 50 from a population of size 1 million tend have 'Benefits' proportion around (fill in here).

Most samples have 'Benefits' proportion between (fill in) and (fill in).

Almost all samples have 'Benefits' proportion between (fill in) and (fill in).

The shape of the distribution of sample proportions is (fill in).

In the same way, one might wonder: if the population is smaller, do we learn much more about it with samples of size '50'?

b. Make a graph of the sampling distribution of sample proportions with **new population size 1000**, and sample size 50. Paste your graph here.

Fill in the blanks:

Simple random samples of size 50 from a population of size 1000 tend have 'Benefits' proportion around (fill in here).

Most samples have 'Benefits' proportion between (fill in) and (fill in).

Almost all samples have 'Benefits' proportion between (fill in) and (fill in).

The shape of the distribution of sample proportions is (fill in).

Group 9

Here are the first two questions from the R document.

1. Imagine that we choose a sample of people from our population and ask people in the sample whether they think that the work scientists do benefits people like them. What population parameter could we use the results of our poll to estimate? What statistic would we calculate to make this estimate?

Parameter: Proportion of people in the world who do believe that the work scientists do benefits them

Statistic: $\hat{p}$, proportion of the sample who believes that scientists work benefits them.

2. If we want to get an unbiased estimate of the population parameter, how should we gather our sample? (For the moment, think about the simplest strategy that we have discussed for creating a representative sample.)

I think that simple random sampling would be best, simply selecting people to participate at random

3. Use the following template to describe the class stacked dot plot.

Simple random samples of size 50 tend have `Benefits` proportion around

78-82

Most samples have `Benefits` proportion between and .

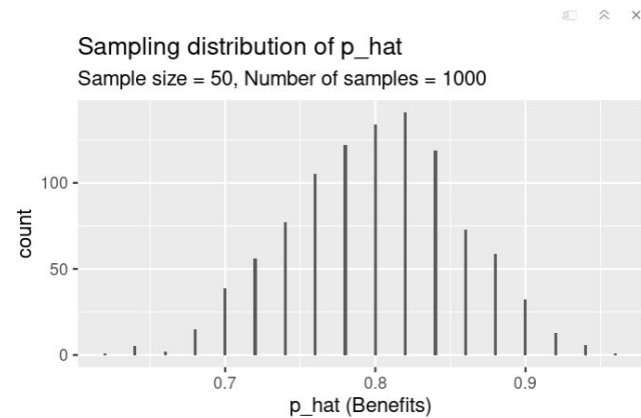
82 and 92 percent

Almost all samples have `Benefits` proportion between and .

75 and 90 percent

The shape of the distribution of sample proportions is slightly left skewed.

4. Put your histogram of the sampling distribution here.



5. Record the mean, standard error, Q1, median, and Q3 for your histogram here.

mean=.799, median=0.8, standard deviation=.05655, Q1: 0.70, Q3: 0.84, standard error=0.00178826801

6. Use the following template to describe the sampling distribution. Use your histogram and the summary statistics from the previous slide.

Simple random samples of size 50 tend have `Benefits` proportion around .799.

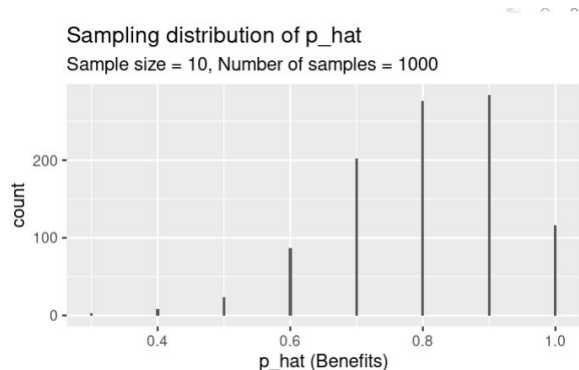
Most samples have `Benefits` proportion between .68 and. 84.

Almost all samples have `Benefits` proportion between .70 and. 90.

The shape of the distribution of sample proportions is normal.

7. What happens when the sample size changes? **Groups on the left-hand side of our classroom answer this one. (If you are on the right-hand side, see the next slide.)**

- a. Create the sampling distribution with samples of size `10` instead of size `50`. Paste your graph below.



Fill in the blanks:

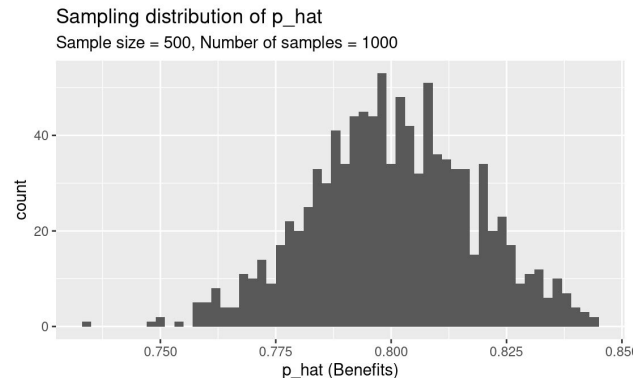
Simple random samples of size 10 tend have `Benefits` proportion around .821.

Most samples have `Benefits` proportion between .7 and .9.

Almost all samples have `Benefits` proportion between .7 and .9.

The shape of the distribution of sample proportions is left skewed.

- b. Create the sampling distribution with samples of size `500` instead of size `50`. Paste your graph below.



Fill in the blanks:

Simple random samples of size 500 tend have `Benefits` proportion around 0.800226.

Most samples have `Benefits` proportion between .788 and .812

Almost all samples have `Benefits` proportion between .764 and .836.

The shape of the distribution of sample proportions is normal bell curve

8. What happens when the population size changes? **Groups on the right-hand side of our classroom answer this one.**

We considered simple random samples drawn from a population of size 100,000. One might wonder: if the population is bigger, do we need to use larger samples to learn about it?

a. Make a graph of sampling distribution of sample proportion with **new population size 1,000,000** and sample size 50. Paste the graph here.

Fill in the blanks:

Simple random samples of size 50 from a population of size 1 million tend have `Benefits` proportion around (fill in here).

Most samples have `Benefits` proportion between (fill in) and (fill in).

Almost all samples have `Benefits` proportion between (fill in) and (fill in).

The shape of the distribution of sample proportions is (fill in).

In the same way, one might wonder: if the population is smaller, do we learn much more about it with samples of size `50`?

b. Make a graph of the sampling distribution of sample proportions with **new population size 1000**, and sample size 50. Paste your graph here.

Fill in the blanks:

Simple random samples of size 50 from a population of size 1000 tend have `Benefits` proportion around (fill in here).

Most samples have `Benefits` proportion between (fill in) and (fill in).

Almost all samples have `Benefits` proportion between (fill in) and (fill in).

The shape of the distribution of sample proportions is (fill in).

Group 10

Here are the first two questions from the R document.

1. Imagine that we choose a sample of people from our population and ask people in the sample whether they think that the work scientists do benefits people like them. What population parameter could we use the results of our poll to estimate? What statistic would we calculate to make this estimate?

Parameter:

Statistic:

2. If we want to get an unbiased estimate of the population parameter, how should we gather our sample? (For the moment, think about the simplest strategy that we have discussed for creating a representative sample.)

3. Use the following template to describe the class stacked dot plot.

Simple random samples of size 50 tend have `Benefits` proportion around

Most samples have `Benefits` proportion between and .

Almost all samples have `Benefits` proportion between and .

The shape of the distribution of sample proportions is slightly.

4. Put your histogram of the sampling distribution here.

5. Record the mean, standard error, Q1, median, and Q3 for your histogram here.

6. Use the following template to describe the sampling distribution. Use your histogram and the summary statistics from the previous slide.

Simple random samples of size 50 tend have `Benefits` proportion around.

Most samples have `Benefits` proportion between and.

Almost all samples have `Benefits` proportion between and.

The shape of the distribution of sample proportions is .

7. What happens when the sample size changes? **Groups on the left-hand side of our classroom answer this one. (If you are on the right-hand side, see the next slide.)**

a. Create the sampling distribution with samples of size `10` instead of size `50`. Paste your graph below.

Fill in the blanks:

Simple random samples of size 10 tend have `Benefits` proportion around.

Most samples have `Benefits` proportion between and.

Almost all samples have `Benefits` proportion between and.

The shape of the distribution of sample proportions is

b. Create the sampling distribution with samples of size `500` instead of size `50`. Paste your graph below.

Fill in the blanks:

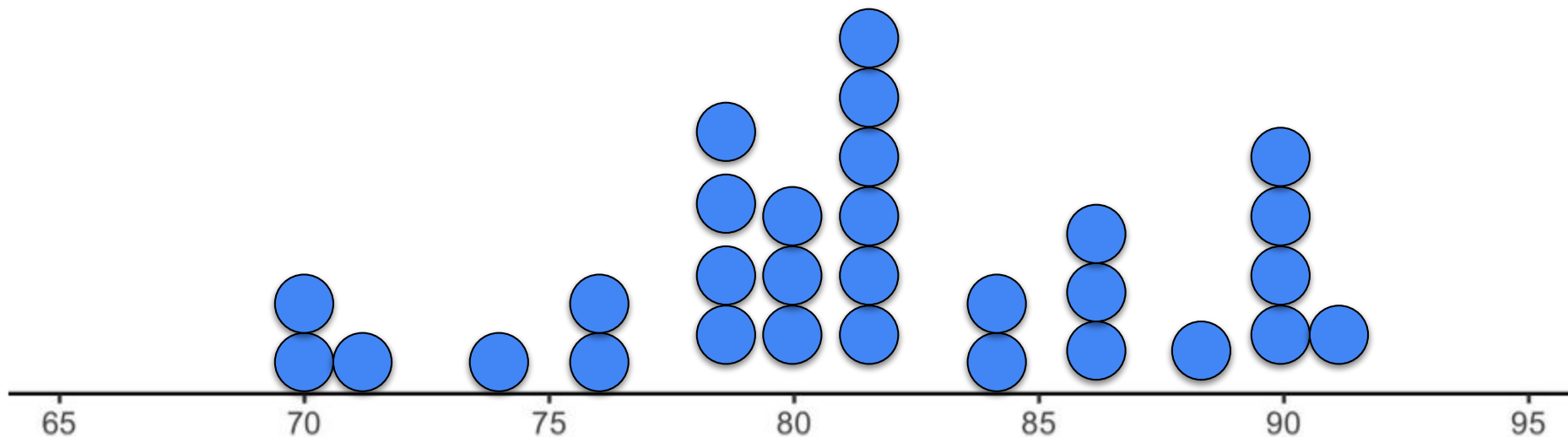
Simple random samples of size 500 tend have `Benefits` proportion around.

Most samples have `Benefits` proportion between and

Almost all samples have `Benefits` proportion between and.

The shape of the distribution of sample proportions is

Class dotplot



```
```{r samp1}  
samp1 <- global_monitor %>%
 slice_sample(n = 50)
```
```

